FreshHarvest
LOGISTICS

Connecting Fresh Produce
from Farms to the World



Prepared by:

O Setshedi	223040891
TM Seepe	222070326
TH Motaung	224227204
FM	224108619
DL Mpapha	224117346

A. Business Case and Existing Database Environment

a) Company, Industry, Products/Services and Main Business Operations

FreshHarvest Logistics is a South African company operating in the agricultural logistics industry. The company specialises in the transportation, storage and distribution of fresh produce such as fruits, vegetables and herbs. Its role is to help move produce efficiently from suppliers and farmers through storage and delivery until it reaches customers.

FreshHarvest provides services such as transport logistics, inventory management, cold storage, shipment tracking, fleet management, supplier coordination, warehouse management and export logistics. Because the products are perishable, accurate information about stock, storage locations, deliveries and customer orders is important to the business.

The main daily operations include receiving produce from suppliers and farmers, updating inventory, processing customer orders, assigning drivers and vehicles to deliveries, tracking shipments, generating invoices and recording customer payments.

Existing database environment. The Semester 1 database was designed as a centralised relational database for these daily operations. It stores information in related tables such as Supplier, Farmer, Batch, Product, ProductCategory, Warehouse, StorageSection, Inventory, Customer, Order, OrderItem, Delivery, Vehicle, Driver, Route, ShipmentTracking, Invoice, Payment, Employee and Equipment. Primary and foreign keys are used to maintain relationships between the tables, and the database was normalised to Third Normal Form (3NF). For the final Business Intelligence implementation, the same operational schema was retained and additional valid historical order transactions were loaded across 2025 and 2026. This provides data across all four quarters and multiple months while preserving the original table relationships and 3NF design.

b) Five Important Operational Transactions

The existing database captures several day-to-day business transactions. Five important transactions are shown below.

No.	Operational transaction	Main tables involved	What is captured
1	Customer places an order	Customer, Order, OrderItem, Product	The customer, order details, products ordered and quantities.
2	Inventory is updated	Product, Inventory, StorageSection, Warehouse	The product, quantity available and where the product is stored.
3	Delivery is created and assigned	Order, Delivery, Driver, Vehicle, Route	The order being delivered, the assigned driver, vehicle and route.
4	Shipment progress is recorded	Delivery, ShipmentTracking	Tracking updates showing the progress or current status of a delivery.
5	Invoice and payment are processed	Order, Invoice, Payment, Employee	Invoice details, payment information and the employee involved in processing the payment.

Why these transactions are important. Together, these transactions represent the main flow of the business. An order records customer demand, inventory shows available stock, delivery records fulfilment, shipment tracking monitors movement, and invoices and payments record the financial side of the transaction.

c) Five Management Questions Requiring Historical, Aggregated or Multidimensional Analysis

The operational database can answer questions about individual orders, products, deliveries or payments. Management, however, also needs summarised information across many transactions and over time. The following management questions require this broader analysis.

No.	Management question	Why broader analysis is needed
1	How do total sales and quantities sold change over time by month, quarter and year?	This requires many orders to be grouped and compared across different time periods.

2	Which products and product categories generate the highest sales value and sales volume?	Management needs totals across many order items rather than viewing one sale at a time.
3	Which customers or customer groups contribute the most to sales over time?	This requires combining multiple orders for each customer and comparing their contribution.
4	How does sales performance differ across delivery routes or logistics areas?	Individual delivery records do not show overall route performance or long-term patterns.
5	How do order or payment statuses affect the value of completed, open or partially settled business?	Status information needs to be aggregated across many orders and invoices to show the overall position.

These questions will guide the next stage of the project because they identify the type of information management wants to analyse. They also show why a Data Warehouse is needed in addition to the existing operational database.

B. Data Warehouse Design

a) Subject Area and Justification

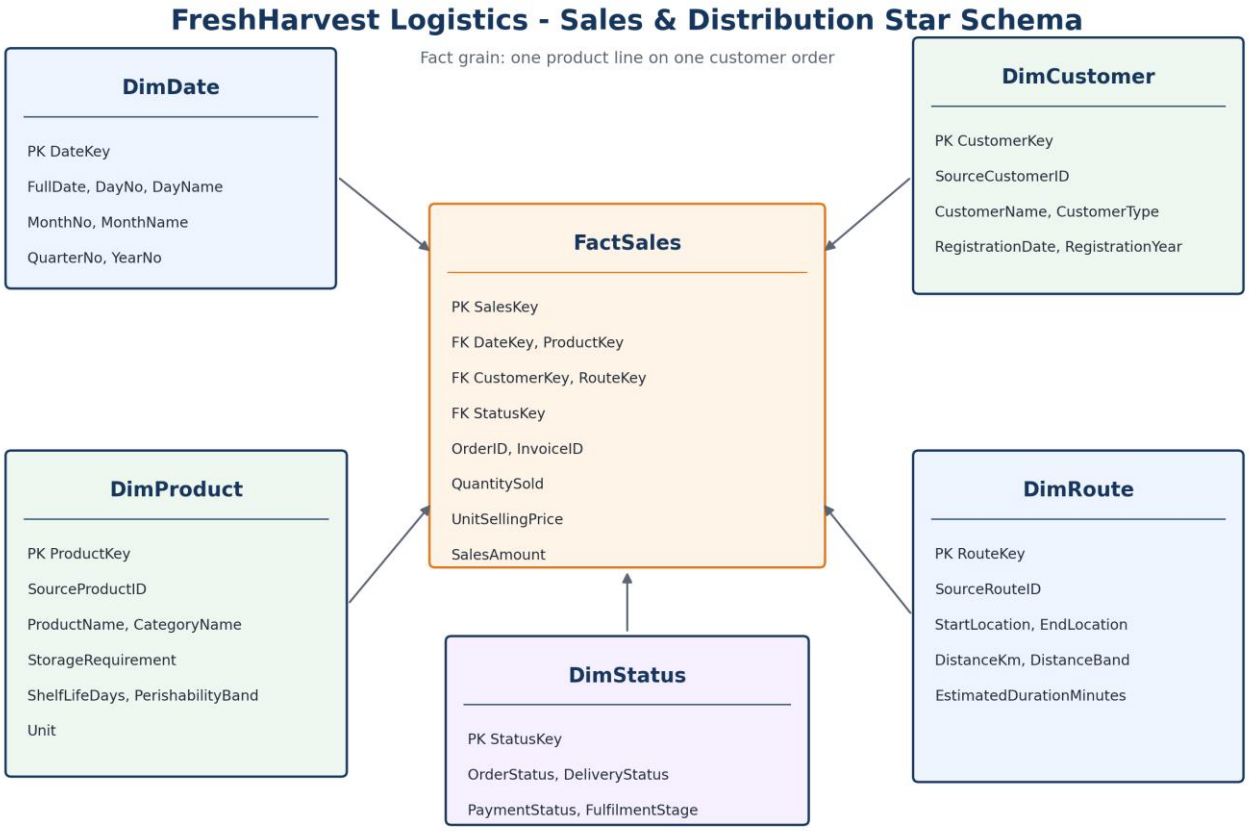
Selected subject area: Sales and Distribution Performance. The proposed Data Warehouse focuses on sales and distribution performance because customer orders connect important parts of FreshHarvest Logistics, including customers, products, order dates, deliveries, routes and transaction statuses.

This subject area directly supports the five management questions identified in Question A. It allows management to analyse sales over time, compare product performance, identify important customers, compare activity across delivery routes and examine the effect of order or payment status on performance.

The existing operational database already contains the source data required for this analysis, especially in the Customer, Order, OrderItem, Product, ProductCategory, Delivery, Route, Invoice and Payment tables. Selected data from these tables is reorganised into the Data Warehouse for reporting and multidimensional analysis.

b) Star Schema Design

The Data Warehouse uses a star schema. The central fact table is FactSales, and it is connected to five dimensions: DimDate, DimProduct, DimCustomer, DimRoute and DimStatus. The grain of FactSales is one product line on one customer order.



Fact Table: FactSales

Attribute	Key/Measure	Description
SalesKey	Primary key	Surrogate identifier for each fact row.
DateKey	Foreign key	Links the sale to DimDate using the order date.
ProductKey	Foreign key	Links the order line to DimProduct.
CustomerKey	Foreign key	Links the order to DimCustomer.
RouteKey	Foreign key	Links the order delivery to DimRoute.
StatusKey	Foreign key	Links order, delivery and payment statuses to DimStatus.
OrderID	Degenerate dimension	Original operational order identifier retained in FactSales.
InvoiceID	Degenerate dimension	Original invoice identifier where available.
QuantitySold	Numerical fact	Quantity from OrderItem.
UnitSellingPrice	Numerical fact	Historical selling price from OrderItem.
SalesAmount	Numerical fact	QuantitySold x UnitSellingPrice.

Dimension Tables

Dimension	Main analytical attributes	Purpose
DimDate	FullDate, DayNo, DayName, MonthNo, MonthName, QuarterNo, YearNo	Time analysis and drill-down/roll-up.
DimProduct	ProductName, CategoryName, StorageRequirement, ShelfLifeDays, PerishabilityBand, Unit	Product and category analysis.
DimCustomer	CustomerName, CustomerType, RegistrationDate, RegistrationYear	Customer and customer-type analysis.
DimRoute	StartLocation, EndLocation, DistanceKm, DistanceBand, EstimatedDurationMinutes	Route, destination and distance analysis.
DimStatus	OrderStatus, DeliveryStatus, PaymentStatus, FulfilmentStage	Combined view of order progress and payment status.

SQL - Dimension and Fact Table Creation

The following SQL is written as editable text so it can be copied directly into SQL Server Management Studio (SSMS).

DimDate

```
CREATE TABLE dbo.DimDate (
    DateKey INT NOT NULL PRIMARY KEY,
    FullDate DATE NOT NULL UNIQUE,
    DayNo TINYINT NOT NULL,
    DayName VARCHAR(10) NOT NULL,
    MonthNo TINYINT NOT NULL,
    MonthName VARCHAR(10) NOT NULL,
    QuarterNo TINYINT NOT NULL,
    YearNo SMALLINT NOT NULL
);
```

DimProduct

```
CREATE TABLE dbo.DimProduct (
    ProductKey INT IDENTITY(1,1) PRIMARY KEY,
    SourceProductID INT NOT NULL UNIQUE,
    ProductName VARCHAR(150) NOT NULL,
    CategoryName VARCHAR(100) NOT NULL,
    StorageRequirement VARCHAR(100) NOT NULL,
    ShelfLifeDays INT NOT NULL,
    PerishabilityBand VARCHAR(30) NOT NULL,
    Unit VARCHAR(20) NOT NULL
);
```

DimCustomer

```
CREATE TABLE dbo.DimCustomer (
    CustomerKey INT IDENTITY(1,1) PRIMARY KEY,
    SourceCustomerID INT NOT NULL UNIQUE,
    CustomerName VARCHAR(150) NOT NULL,
    CustomerType VARCHAR(50) NOT NULL,
    RegistrationDate DATE NOT NULL,
    RegistrationYear SMALLINT NOT NULL
);
```

DimRoute

```
CREATE TABLE dbo.DimRoute (
    RouteKey INT IDENTITY(1,1) PRIMARY KEY,
    SourceRouteID INT NULL UNIQUE,
    StartLocation VARCHAR(150) NOT NULL,
    EndLocation VARCHAR(150) NOT NULL,
    DistanceKm DECIMAL(8,2) NOT NULL,
    DistanceBand VARCHAR(30) NOT NULL,
    EstimatedDurationMinutes INT NOT NULL
);
```

DimStatus

```
CREATE TABLE dbo.DimStatus (
    StatusKey INT IDENTITY(1,1) PRIMARY KEY,
    OrderStatus VARCHAR(30) NOT NULL,
    DeliveryStatus VARCHAR(30) NOT NULL,
    PaymentStatus VARCHAR(30) NOT NULL,
    FulfilmentStage VARCHAR(30) NOT NULL,
    CONSTRAINT UQ_DimStatus
        UNIQUE (OrderStatus, DeliveryStatus, PaymentStatus)
);
```

FactSales

```
CREATE TABLE dbo.FactSales (
    SalesKey BIGINT IDENTITY(1,1) PRIMARY KEY,
    DateKey INT NOT NULL,
    ProductKey INT NOT NULL,
    CustomerKey INT NOT NULL,
    RouteKey INT NOT NULL,
    StatusKey INT NOT NULL,
    OrderID INT NOT NULL,
    InvoiceID INT NULL,
    QuantitySold DECIMAL(10,2) NOT NULL,
    UnitSellingPrice DECIMAL(10,2) NOT NULL,
    SalesAmount DECIMAL(14,2) NOT NULL,
    FOREIGN KEY (DateKey) REFERENCES dbo.DimDate(DateKey),
    FOREIGN KEY (ProductKey) REFERENCES dbo.DimProduct(ProductKey),
    FOREIGN KEY (CustomerKey) REFERENCES dbo.DimCustomer(CustomerKey),
    FOREIGN KEY (RouteKey) REFERENCES dbo.DimRoute(RouteKey),
    FOREIGN KEY (StatusKey) REFERENCES dbo.DimStatus(StatusKey)
);
```

c) Operational Database to Data Warehouse Mapping

The Data Warehouse is loaded from the existing Semester 1 Fresh_Harvest operational database. The mapping below shows where each warehouse table gets its data.

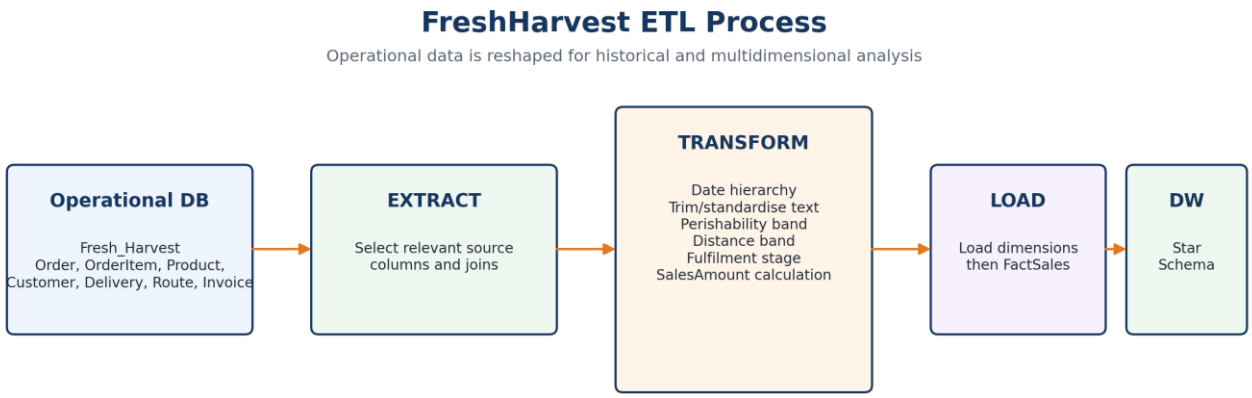
Warehouse target	Operational source	Source attributes	Transformation / mapping
DimDate	Order	OrderDate	Create DateKey and derive day, month, quarter and year.
DimProduct	Product + ProductCategory	ProductID, ProductName, ShelfLifeDays, Unit, CategoryName, StorageRequirement	Join category data to product, trim text and derive PerishabilityBand.
DimCustomer	Customer	CustomerID, CustomerName, CustomerType, RegistrationDate	Trim text and derive RegistrationYear.
DimRoute	Route	RouteID, StartLocation, EndLocation, Distance, EstimatedDurationMinutes	Rename Distance to DistanceKm, derive DistanceBand and add an Unknown route member.
DimStatus	Order + Delivery + Invoice	Order.Status, Delivery.Status, Invoice.PaymentStatus	Combine three status sources and derive FulfilmentStage.
FactSales	Order + OrderItem + Delivery + Route + Invoice	OrderID, OrderDate, CustomerID, ProductID, Quantity, UnitPrice, RouteID, InvoiceID	Look up dimension keys and calculate SalesAmount = Quantity x UnitPrice.

d) Data Warehouse Characteristics

Characteristic	How the proposed Data Warehouse demonstrates it
Subject-oriented	The warehouse focuses on Sales and Distribution Performance instead of separate operational activities.

Characteristic	How the proposed Data Warehouse demonstrates it
Integrated	Customer, product, category, route, delivery and invoice data from different Semester 1 tables is combined into consistent dimensions and one fact table.
Time-variant	DimDate stores day, month, quarter and year values, and every FactSales row is linked to a DateKey.
Nonvolatile	The warehouse is used for reporting and analysis. Historical fact rows are queried and aggregated rather than updated as normal daily transactions.

e) Extract, Transform and Load (ETL) Process



The ETL process extracts relevant source data from Fresh_Harvest, transforms it into analysis-ready values and then loads the dimensions before FactSales. Loading dimensions first allows the fact table to use valid surrogate foreign keys.

Meaningful Transformations

Transformation	Source example	Warehouse result	Reason
T1 - Date hierarchy	OrderDate = 2026-05-05	DateKey 20260505; May; Q2; 2026	Supports analysis by day, month, quarter and year.
T2 - Source integration	Product = Spinach; Category = Leafy Greens	One DimProduct row with product and category data	Reduces repeated joins during analysis.
T3 - Perishability classification	Spinach ShelfLifeDays = 5	Highly Perishable	Turns shelf-life days into a useful analytical category.
T4 - Route classification	Pretoria Cold Storage to Cape Town Harbour; 1450 km	Long Distance (>500 km)	Allows route performance comparison by distance band.
T5 - Fulfilment stage	Processing + In Transit + Partially Paid	In Transit	Combines three operational status fields.
T6 - Sales calculation	100 x R18.50	R1,850.00	Creates the main revenue measure.
T7 - Unknown route handling	Order without an assigned route	DimRoute = Unknown	Protects fact-table referential integrity.

ETL SQL

1. Load DimDate

```
INSERT INTO dbo.DimDate
    (DateKey, FullDate, DayNo, DayName,
     MonthNo, MonthName, QuarterNo, YearNo)
SELECT DISTINCT
    CONVERT(INT, CONVERT(CHAR(8), CAST(o.OrderDate AS DATE), 112)),
    CAST(o.OrderDate AS DATE),
    DAY(o.OrderDate),
    DATENAME(WEEKDAY, o.OrderDate),
    MONTH(o.OrderDate),
    DATENAME(MONTH, o.OrderDate),
    DATEPART(QUARTER, o.OrderDate),
    YEAR(o.OrderDate)
FROM Fresh_Harvest.dbo.[Order] o;
```

2. Load DimProduct

```
INSERT INTO dbo.DimProduct
    (SourceProductID, ProductName, CategoryName,
     StorageRequirement, ShelfLifeDays,
     PerishabilityBand, Unit)
SELECT
    p.ProductID,
    LTRIM(RTRIM(p.ProductName)),
    LTRIM(RTRIM(pc.CategoryName)),
    LTRIM(RTRIM(pc.StorageRequirement)),
    p.ShelfLifeDays,
    CASE
        WHEN p.ShelfLifeDays <= 5 THEN 'Highly Perishable'
        WHEN p.ShelfLifeDays <= 14 THEN 'Perishable'
        ELSE 'Extended Shelf Life'
    END,
    LTRIM(RTRIM(p.Unit))
FROM Fresh_Harvest.dbo.Product p
JOIN Fresh_Harvest.dbo.ProductCategory pc
    ON pc.CategoryID = p.CategoryID;
```

3. Load DimCustomer

```
INSERT INTO dbo.DimCustomer
    (SourceCustomerID, CustomerName, CustomerType,
     RegistrationDate, RegistrationYear)
SELECT
    c.CustomerID,
    LTRIM(RTRIM(c.CustomerName)),
    LTRIM(RTRIM(c.CustomerType)),
    c.RegistrationDate,
    YEAR(c.RegistrationDate)
FROM Fresh_Harvest.dbo.Customer c;
```

4. Load DimRoute

```
INSERT INTO dbo.DimRoute
    (SourceRouteID, StartLocation, EndLocation,
     DistanceKm, DistanceBand, EstimatedDurationMinutes)
SELECT
    r.RouteID,
    LTRIM(RTRIM(r.StartLocation)),
    LTRIM(RTRIM(r.EndLocation)),
    r.Distance,
    CASE
        WHEN r.Distance <= 50 THEN 'Local (0-50 km)'
        WHEN r.Distance <= 500 THEN 'Regional (51-500 km)'
        ELSE 'Long Distance (>500 km)'
    END,
    r.EstimatedDurationMinutes
FROM Fresh_Harvest.dbo.Route r;

INSERT INTO dbo.DimRoute
```

```
    (SourceRouteID, StartLocation, EndLocation,
     DistanceKm, DistanceBand, EstimatedDurationMinutes)
VALUES (NULL, 'Unknown', 'Unknown', 0, 'Unknown', 0);
```

5. Load DimStatus

```
INSERT INTO dbo.DimStatus
    (OrderStatus, DeliveryStatus,
     PaymentStatus, FulfilmentStage)
SELECT DISTINCT
    COALESCE(o.Status, 'Unknown'),
    COALESCE(d.Status, 'Unknown'),
    COALESCE(i.PaymentStatus, 'Unknown'),
    CASE
        WHEN o.Status = 'Cancelled' OR d.Status = 'Cancelled'
            THEN 'Cancelled'
        WHEN d.Status = 'Delivered' AND i.PaymentStatus = 'Paid'
            THEN 'Completed'
        WHEN d.Status = 'Delivered'
            THEN 'Delivered - Payment Open'
        WHEN d.Status = 'In Transit' THEN 'In Transit'
        WHEN d.Status = 'Scheduled' THEN 'Scheduled'
        WHEN o.Status = 'Processing' THEN 'Processing'
        WHEN o.Status = 'Pending' THEN 'Pending'
        ELSE COALESCE(o.Status, 'Unknown')
    END
FROM Fresh_Harvest.dbo.[Order] o
LEFT JOIN Fresh_Harvest.dbo.Delivery d
    ON d.OrderID = o.OrderID
LEFT JOIN Fresh_Harvest.dbo.Invoice i
    ON i.OrderID = o.OrderID;
```

6. Load FactSales

```
INSERT INTO dbo.FactSales
    (DateKey, ProductKey, CustomerKey, RouteKey, StatusKey,
     OrderID, InvoiceID, QuantitySold,
     UnitSellingPrice, SalesAmount)
SELECT
    dd.DateKey,
    dp.ProductKey,
    dc.CustomerKey,
    COALESCE(dr.RouteKey, ur.RouteKey),
    ds.StatusKey,
    o.OrderID,
    i.InvoiceID,
    oi.Quantity,
    oi.UnitPrice,
    CAST(oi.Quantity * oi.UnitPrice AS DECIMAL(14,2))
FROM Fresh_Harvest.dbo.[Order] o
JOIN Fresh_Harvest.dbo.OrderItem oi
    ON oi.OrderID = o.OrderID
JOIN dbo.DimDate dd
    ON dd.FullDate = CAST(o.OrderDate AS DATE)
JOIN dbo.DimProduct dp
    ON dp.SourceProductID = oi.ProductID
JOIN dbo.DimCustomer dc
    ON dc.SourceCustomerID = o.CustomerID
LEFT JOIN Fresh_Harvest.dbo.Delivery d
    ON d.OrderID = o.OrderID
LEFT JOIN Fresh_Harvest.dbo.Route r
    ON r.RouteID = d.RouteID
LEFT JOIN dbo.DimRoute dr
    ON dr.SourceRouteID = r.RouteID
CROSS APPLY (
    SELECT TOP 1 RouteKey
    FROM dbo.DimRoute
    WHERE SourceRouteID IS NULL
) ur
LEFT JOIN Fresh_Harvest.dbo.Invoice i
```

```
    ON i.OrderID = o.OrderID
JOIN dbo.DimStatus ds
    ON ds.OrderStatus = COALESCE(o.Status, 'Unknown')
   AND ds.DeliveryStatus = COALESCE(d.Status, 'Unknown')
   AND ds.PaymentStatus = COALESCE(i.PaymentStatus, 'Unknown');
```

f) Evidence of Successful Loading

After the historical source transactions are loaded and the ETL is executed again, the final Data Warehouse contains data for 22 customer orders across 2025 and 2026. The final warehouse supports analysis by year, quarter and month and contains the following row counts.

Warehouse table	Rows loaded
DimDate	22
DimProduct	5
DimCustomer	3
DimRoute	4 (3 source routes + 1 Unknown)
DimStatus	8
FactSales	41

Load-count SQL

```
SELECT 'DimDate' AS TableName, COUNT(*) AS RowCount FROM dbo.DimDate
UNION ALL SELECT 'DimProduct', COUNT(*) FROM dbo.DimProduct
UNION ALL SELECT 'DimCustomer', COUNT(*) FROM dbo.DimCustomer
UNION ALL SELECT 'DimRoute', COUNT(*) FROM dbo.DimRoute
UNION ALL SELECT 'DimStatus', COUNT(*) FROM dbo.DimStatus
UNION ALL SELECT 'FactSales', COUNT(*) FROM dbo.FactSales;
```

Final warehouse validation summary

Validation measure	Final result
Distinct orders	22
Distinct order dates	22
Total FactSales rows	41
Total quantity sold	6,610
Total pre-VAT sales revenue	R115,440.00
Years represented	2025 and 2026
Quarter coverage	Q1, Q2, Q3 and Q4

Referential-integrity validation

```
SELECT
    SUM(CASE WHEN dd.DateKey IS NULL THEN 1 ELSE 0 END) AS MissingDateKeys,
    SUM(CASE WHEN dp.ProductKey IS NULL THEN 1 ELSE 0 END) AS MissingProductKeys,
    SUM(CASE WHEN dc.CustomerKey IS NULL THEN 1 ELSE 0 END) AS MissingCustomerKeys,
    SUM(CASE WHEN dr.RouteKey IS NULL THEN 1 ELSE 0 END) AS MissingRouteKeys,
    SUM(CASE WHEN ds.StatusKey IS NULL THEN 1 ELSE 0 END) AS MissingStatusKeys
FROM dbo.FactSales fs
LEFT JOIN dbo.DimDate dd ON dd.DateKey = fs.DateKey
LEFT JOIN dbo.DimProduct dp ON dp.ProductKey = fs.ProductKey
LEFT JOIN dbo.DimCustomer dc ON dc.CustomerKey = fs.CustomerKey
LEFT JOIN dbo.DimRoute dr ON dr.RouteKey = fs.RouteKey
LEFT JOIN dbo.DimStatus ds ON ds.StatusKey = fs.StatusKey;
```

Expected result: all five missing-key values are 0. The final FactSales dataset contains 41 order-line rows from 22 distinct orders, with 6,610 units sold and R115,440.00 in pre-VAT sales revenue.

C. SQL Queries and OLAP Operations

The following queries use the implemented star schema. Each query includes the business question, editable SQL, the result from the final historical dataset and a short explanation of the OLAP operation.

C1. Slice

Business question: During May 2026, how much sales revenue did each product category generate?

```
SELECT
    dp.CategoryName,
    SUM(fs.SalesAmount) AS SalesRevenue
FROM dbo.FactSales fs
JOIN dbo.DimDate dd ON dd.DateKey = fs.DateKey
JOIN dbo.DimProduct dp ON dp.ProductKey = fs.ProductKey
WHERE dd.YearNo = 2026
  AND dd.MonthNo = 5
GROUP BY dp.CategoryName
ORDER BY SalesRevenue DESC;
```

Product category	Sales revenue
Citrus Fruits	R8,800.00
Root Vegetables	R2,500.00
Leafy Greens	R1,850.00
Fresh Herbs	R500.00

OLAP explanation: This is a Slice because one time slice, May 2026, is selected from the Date dimension while revenue is analysed by product category.

C2. Dice

Business question: In 2026, what revenue was generated by Retailer and Exporter customers for Citrus Fruits and Leafy Greens?

```
SELECT
    dc.CustomerType,
    dp.CategoryName,
    SUM(fs.SalesAmount) AS SalesRevenue
FROM dbo.FactSales fs
JOIN dbo.DimDate dd ON dd.DateKey = fs.DateKey
JOIN dbo.DimCustomer dc ON dc.CustomerKey = fs.CustomerKey
JOIN dbo.DimProduct dp ON dp.ProductKey = fs.ProductKey
WHERE dd.YearNo = 2026
  AND dc.CustomerType IN ('Retailer','Exporter')
  AND dp.CategoryName IN ('Citrus Fruits','Leafy Greens')
GROUP BY dc.CustomerType, dp.CategoryName
ORDER BY dc.CustomerType, SalesRevenue DESC;
```

Customer type	Product category	Sales revenue
Exporter	Citrus Fruits	R27,500.00
Retailer	Citrus Fruits	R9,680.00
Retailer	Leafy Greens	R12,820.00

OLAP explanation: This is a Dice because several dimension conditions are applied together: Year = 2026, CustomerType in {Retailer, Exporter}, and ProductCategory in {Citrus Fruits, Leafy Greens}.

C3. Drill-down

Business question: After viewing May 2026 sales at month level, what were the sales values on each individual day?

```
SELECT
    dd.YearNo,
    dd.MonthName,
    dd.FullDate,
    SUM(fs.SalesAmount) AS DailySalesRevenue
FROM dbo.FactSales fs
JOIN dbo.DimDate dd ON dd.DateKey = fs.DateKey
WHERE dd.YearNo = 2026
  AND dd.MonthNo = 5
GROUP BY dd.YearNo, dd.MonthName, dd.FullDate
ORDER BY dd.FullDate;
```

Year	Month	Date	Daily sales revenue
2026	May	05 May 2026	R5,150.00
2026	May	06 May 2026	R3,000.00
2026	May	07 May 2026	R5,500.00

OLAP explanation: Drill-down moves from the higher month level of the time hierarchy to the more detailed individual-date level.

C4. Roll-up

Business question: What are the detailed product sales, product-category subtotals and overall sales total?

```
SELECT
    CASE WHEN GROUPING(dp.CategoryName) = 1
         THEN 'ALL CATEGORIES'
         ELSE dp.CategoryName END AS CategoryName,
    CASE
        WHEN GROUPING(dp.CategoryName) = 1 THEN 'GRAND TOTAL'
        WHEN GROUPING(dp.ProductName) = 1 THEN 'CATEGORY TOTAL'
        ELSE dp.ProductName
    END AS ProductLevel,
    SUM(fs.SalesAmount) AS SalesRevenue
FROM dbo.FactSales fs
JOIN dbo.DimProduct dp ON dp.ProductKey = fs.ProductKey
GROUP BY ROLLUP(dp.CategoryName, dp.ProductName);
```

Category	Product / level	Sales revenue
Citrus Fruits	Oranges	R62,480.00
Citrus Fruits	CATEGORY TOTAL	R62,480.00
Fresh Herbs	Coriander	R4,000.00
Fresh Herbs	CATEGORY TOTAL	R4,000.00
Leafy Greens	Lettuce	R8,250.00
Leafy Greens	Spinach	R16,835.00
Leafy Greens	CATEGORY TOTAL	R25,085.00
Root Vegetables	Potatoes	R23,875.00
Root Vegetables	CATEGORY TOTAL	R23,875.00
ALL CATEGORIES	GRAND TOTAL	R115,440.00

OLAP explanation: Roll-up aggregates detailed product rows into product-category totals and then into the overall grand total.

C5. Pivot

Business question: How is product-category revenue distributed across Retailer, Wholesaler and Exporter customer types?

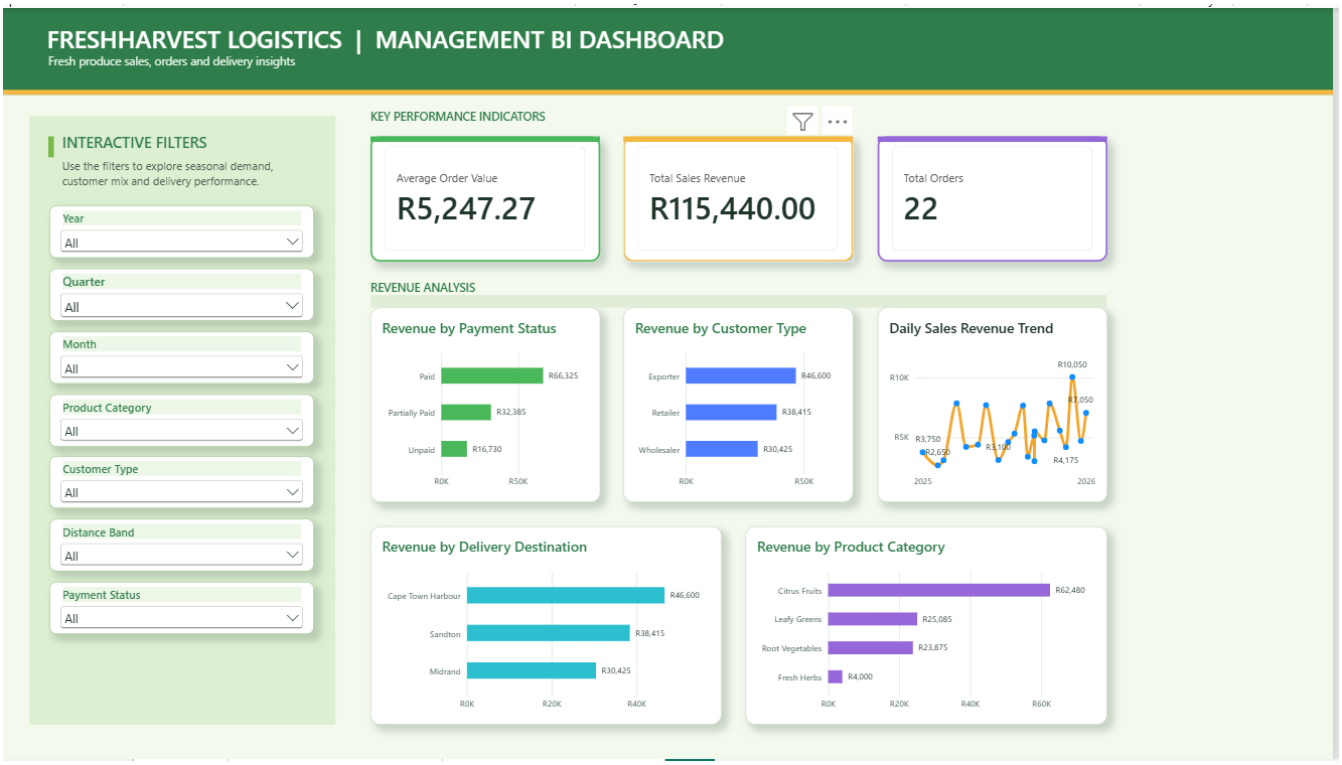
```
SELECT
    CategoryName,
    ISNULL([Retailer], 0) AS Retailer,
    ISNULL([Wholesaler], 0) AS Wholesaler,
    ISNULL([Exporter], 0) AS Exporter
FROM (
    SELECT
        dp.CategoryName,
        dc.CustomerType,
        fs.SalesAmount
    FROM dbo.FactSales fs
    JOIN dbo.DimProduct dp
        ON dp.ProductKey = fs.ProductKey
    JOIN dbo.DimCustomer dc
        ON dc.CustomerKey = fs.CustomerKey
) src
PIVOT (
    SUM(SalesAmount)
    FOR CustomerType IN ([Retailer], [Wholesaler], [Exporter])
) p
ORDER BY CategoryName;
```

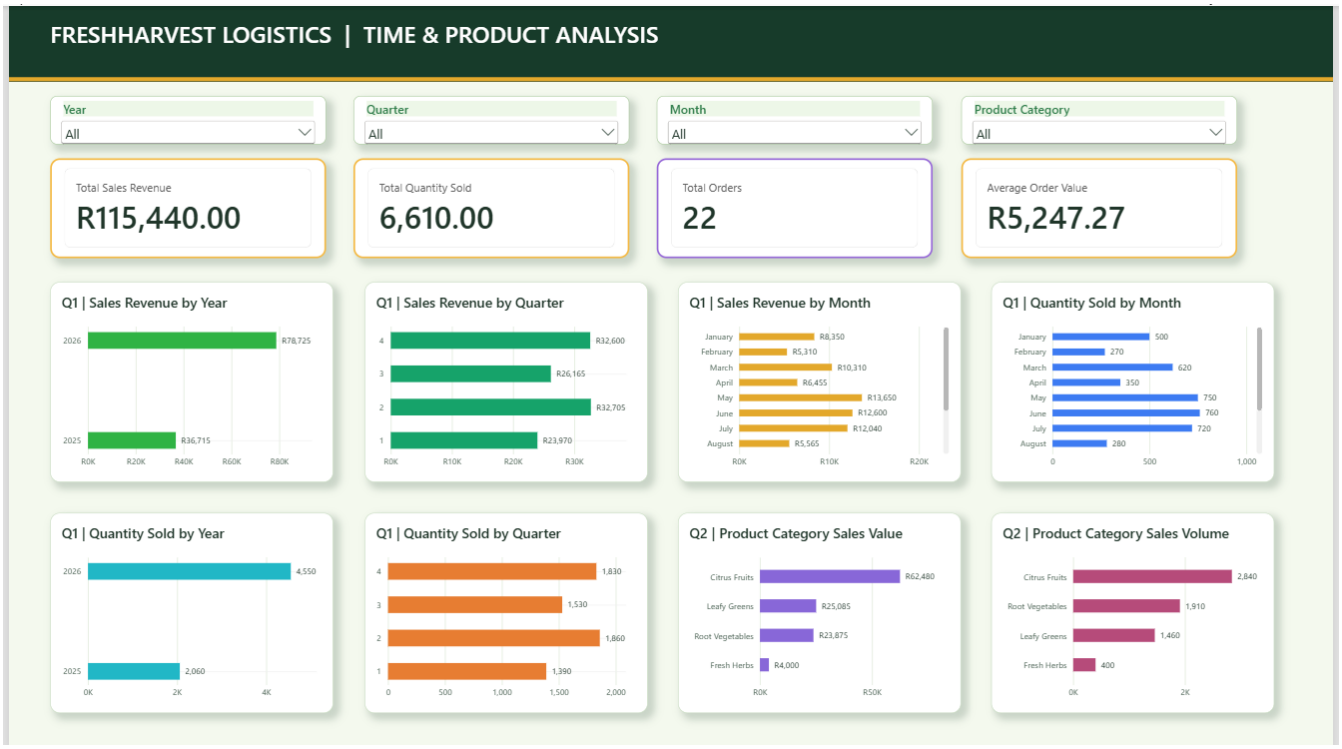
Product category	Retailer	Wholesaler	Exporter
Citrus Fruits	R17,380.00	R3,300.00	R41,800.00
Fresh Herbs	R2,300.00	R900.00	R800.00
Leafy Greens	R18,735.00	R6,350.00	R0.00
Root Vegetables	R0.00	R19,875.00	R4,000.00

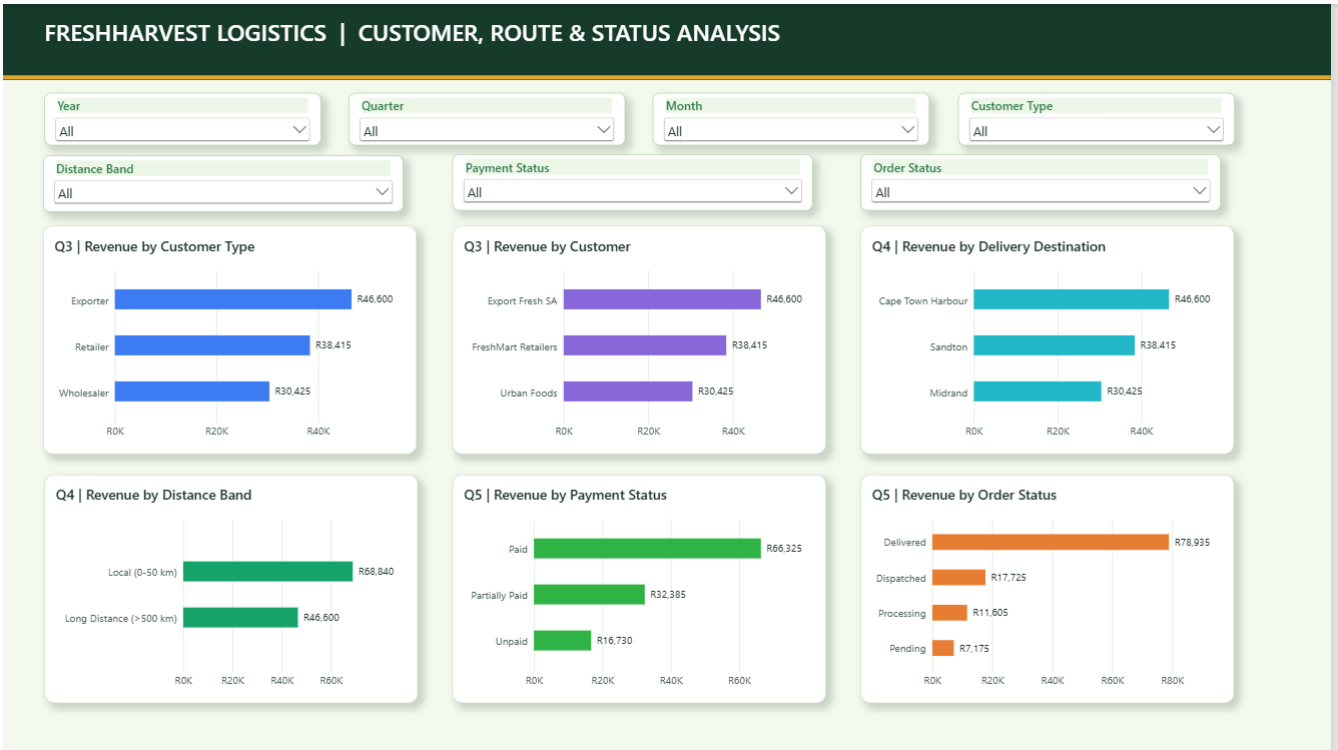
OLAP explanation: Pivot changes the presentation orientation by turning CustomerType values into columns. This makes it easy to compare customer segments side by side for each product category.

D. Business Intelligence Dashboard and Decision Support

The final Business Intelligence layer was implemented in Microsoft Power BI using the FreshHarvest_DW star schema. FactSales provides sales revenue, quantity and order measures, while DimDate, DimProduct, DimCustomer, DimRoute and DimStatus provide the dimensions used for filtering and multidimensional analysis. Revenue values are based on the pre-VAT SalesAmount loaded from OrderItem.Quantity x OrderItem.UnitPrice. The final report uses historical data from 2025 and 2026 and is organised into three connected analytical pages so that all five management questions can be answered.







D(a). Dashboard Visualisations

The final Power BI report is organised into three report pages: Management BI Dashboard, Time & Product Analysis, and Customer, Route & Status Analysis. Together they provide revenue trends, quantity analysis, product performance, customer contribution, route analysis and status analysis. All visuals use the same Data Warehouse relationships so slicer selections recalculate the relevant KPI cards and charts.

Visualisation / analysis	Main fields / measure	Management purpose
Sales Revenue by Year, Quarter and Month	DimDate[YearNo], DimDate[QuarterNo], DimDate[MonthName], [Total Sales Revenue]	Compares revenue across 2025 and 2026 and supports year, quarter and month analysis.
Quantity Sold by Year, Quarter and Month	DimDate time hierarchy and [Total Quantity Sold]	Shows how sales volume changes across different time periods.
Product Category Sales Value	DimProduct[CategoryName] and [Total Sales Revenue]	Identifies the product categories generating the highest monetary value.
Product Category Sales Volume	DimProduct[CategoryName] and [Total Quantity Sold]	Identifies the categories selling the greatest quantity.
Revenue by Customer Type and Customer	DimCustomer[CustomerType], DimCustomer[CustomerName], [Total Sales Revenue]	Shows which customer groups and individual customers contribute most to sales.
Revenue by Delivery Destination and Distance Band	DimRoute[EndLocation], DimRoute[DistanceBand], [Total Sales Revenue]	Compares sales performance across logistics areas and local or long-distance routes.
Revenue by Payment Status and Order Status	DimStatus[PaymentStatus], DimStatus[OrderStatus], [Total Sales Revenue]	Shows the value of paid, partially paid, unpaid, delivered, dispatched, processing and pending business.
Daily Sales Revenue Trend	DimDate[FullDate] and [Total Sales Revenue]	Shows daily fluctuations and supports detailed time-based monitoring.

D(b). Key Performance Indicators

Four KPI cards are used across the analytical report to provide management with an immediate summary of sales value, sales volume, order activity and average order value.

KPI	Calculation	Final value
Total Sales Revenue	SUM(FactSales[SalesAmount])	R115,440.00
Total Quantity Sold	SUM(FactSales[QuantitySold])	6,610
Total Orders	DISTINCTCOUNT(FactSales[OrderID])	22
Average Order Value	[Total Sales Revenue] / [Total Orders]	R5,247.27

Power BI DAX measures used:

```
Total Sales Revenue = SUM(FactSales[SalesAmount])
Total Quantity Sold = SUM(FactSales[QuantitySold])

Total Orders = DISTINCTCOUNT(FactSales[OrderID])
Average Order Value = DIVIDE([Total Sales Revenue], [Total Orders])
```

D(c). Filtering Capabilities

The final report provides interactive slicers for Year, Quarter, Month, Product Category, Customer Type, Distance Band, Payment Status and Order Status. These filters allow management to move from broad historical analysis to a specific period, product, customer group, logistics area or status. Selecting a slicer automatically updates all linked visuals and KPI cards on the relevant report page.

D(d). Answers to the Five Management Questions

The final Power BI report answers the five management questions identified at the beginning of the project using the expanded 2025-2026 warehouse data.

Management question	Answer from the final Power BI report
1. How do total sales and quantities sold change over time by month, quarter and year?	Revenue increased from R36,715.00 in 2025 to R78,725.00 in 2026. Quantity sold increased from 2,060 to 4,550. Across both years, Q2 produced the highest revenue at R32,705.00, closely followed by Q4 at R32,600.00. In 2026, May was the strongest month at R13,650.00.
2. Which products and product categories generate the highest sales value and sales volume?	Citrus Fruits is the leading category for both value and volume, generating R62,480.00 and 2,840 units. Leafy Greens generated R25,085.00, Root Vegetables R23,875.00 and Fresh Herbs R4,000.00.
3. Which customers or customer groups contribute the most to sales over time?	Exporter customers contribute the most revenue at R46,600.00. Retailers contribute R38,415.00 and Wholesalers R30,425.00. Export Fresh SA is the highest individual customer at R46,600.00, followed by FreshMart Retailers at R38,415.00 and Urban Foods at R30,425.00.
4. How does sales performance differ across delivery routes or logistics areas?	Cape Town Harbour is the highest individual delivery destination at R46,600.00, followed by Sandton at R38,415.00 and Midrand at R30,425.00. Local routes generate R68,840.00 in total compared with R46,600.00 from long-distance routes.
5. How do order or payment statuses affect completed, open or partially settled business?	Paid business contributes R66,325.00, Partially Paid R32,385.00 and Unpaid R16,730.00. By order status, Delivered contributes R78,935.00, Dispatched R17,725.00, Processing R11,605.00 and Pending R7,175.00.

D(e). Important Insights and Supported Management Decisions

The final historical dataset provides management with stronger evidence for decision support because the analysis now covers two years, all four quarters and multiple operational statuses.

Insight from the final dataset	Management decision supported
Citrus Fruits generate R62,480.00 of R115,440.00 total revenue (approximately 54.1%) and also have the highest sales volume at 2,840 units.	Management can prioritise citrus procurement, storage capacity and stock availability because this category is the strongest contributor by both value and volume.

Exporter customers contribute R46,600.00 (approximately 40.4% of total revenue), and the same amount is associated with the Cape Town Harbour long-distance destination.	Management can monitor export-route capacity, vehicle availability and long-distance logistics costs because export business represents a major share of revenue.
Paid business represents R66,325.00, while R49,115.00 is either partially paid or unpaid.	Management can use the payment-status analysis to prioritise collection follow-up and monitor cash-flow exposure while distinguishing settled from outstanding business.

Power BI Dashboard Design Rationale

The dashboard uses a farm-inspired visual identity that reflects the FreshHarvest business context while remaining clean and readable for management. A dark green header and fresh light-green background are combined with white cards and clear green, blue, purple, orange and gold accents. Large KPI values, consistent chart spacing and grouped filters improve readability during both normal use and live presentation. The three-page layout separates high-level management monitoring from detailed time/product analysis and customer/route/status analysis.

Conclusion

FreshHarvest Logistics has been extended from its Semester 1 operational database into a complete Data Warehouse and Business Intelligence solution. The operational schema now contains historical transactions across 2025 and 2026, while the implemented star schema integrates time, product, customer, route and status information around FactSales. The ETL process loads 41 fact rows from 22 orders and supports historical analysis across year, quarter and month. The OLAP queries demonstrate Slice, Dice, Drill-down, Roll-up and Pivot operations using the final warehouse data.

The final Power BI report completes the analytical solution with three interactive pages and four KPI measures. It answers all five management questions by analysing revenue and quantity over time, product value and volume, customer contribution, delivery-route performance, and payment and order status. Together, SQL Server, the FreshHarvest_DW star schema, OLAP operations and Power BI provide a complete decision-support environment based on R115,440.00 of historical sales revenue, 6,610 units sold and 22 orders.